

ATTACHMENT G

GOVERNMENT FURNISHED EQUIPMENT (GFE)

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CONTRACT TBD

Attachment G - Government Furnished Equipment (GFE)

This document has been reviewed and determined not to contain export-controlled technical information.

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Introduction

NASA has identified the need for establishing a robust Mars relay infrastructure to provide mission-enabling and enhancing telecommunications and navigation services to future exploration elements. To this end, NASA has developed the Electra UHF Transceiver (EUT). The EUT is shown in Figure 1. Redundant EUTs were flown for the first time on the 2005 Mars Reconnaissance Orbiter (MRO) mission. A single-string Electra has since been launched onboard the 2013 Mars Atmosphere and Volatile Evolution Mission (MAVEN), and a dual-string version has been launched on the 2016 European Space Agency ExoMars/Trace Gas Orbiter. In addition, transceivers compatible with the EUT have flown on Phoenix, MSL, Insight, and Mars 2020. In recent years, the EUT has been upgraded to both address obsolete parts and add additional capabilities.

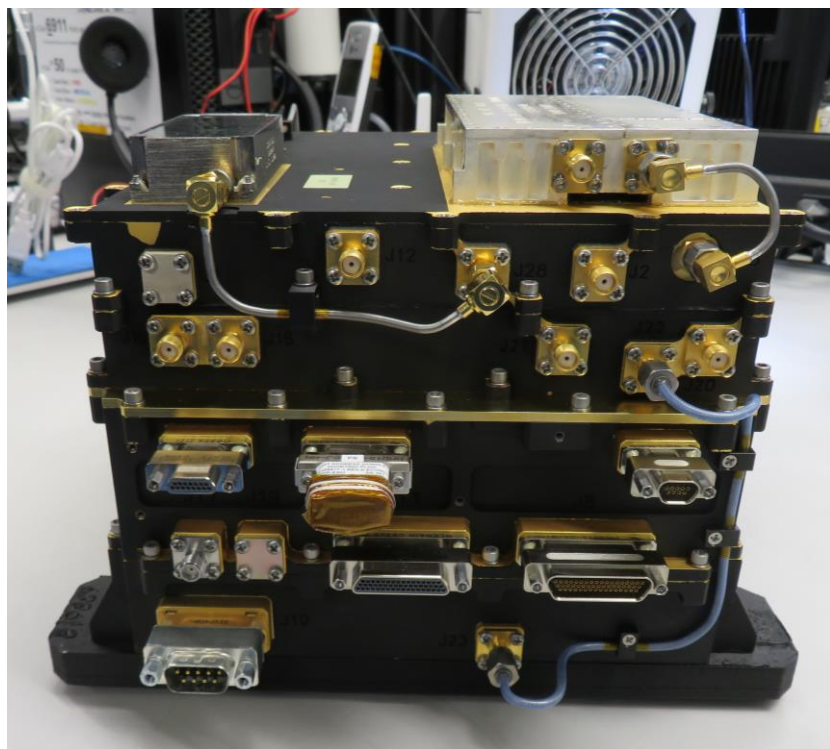


Figure 1: Electra UHF Transceiver

As a continuation of this strategy, NASA is providing two Flight Model (FM) units of the EUT radio for use on the 2028 Mars Telecommunications Network (MTN) Orbiter. The transceiver and ground support will be provided by NASA. However, the Contractor is responsible for any and all costs associated with integrating the transceiver into the overall flight system, and for implementing the functional capability within the project mission operation system/ground data system required to operate the transceiver. Deliverables to the selected MTN Contractor will consist of two EUT FM units and a

single baseband-only prototype with flight-like digital interfaces. For higher-fidelity compatibility testing an Engineering Model (EM) EUT will be provided to the selected MTN provider for test and return. Additionally, to support integration and testing of the EM and/or FM units with the larger flight system, a set of Ground Support Equipment (GSE) capable of serving as UHF relay partner will be provided to the selected MTN provider for test and return. Schedule and durations of loaned equipment will be determined by the Government based on the Contractor's schedule and Agency priorities.

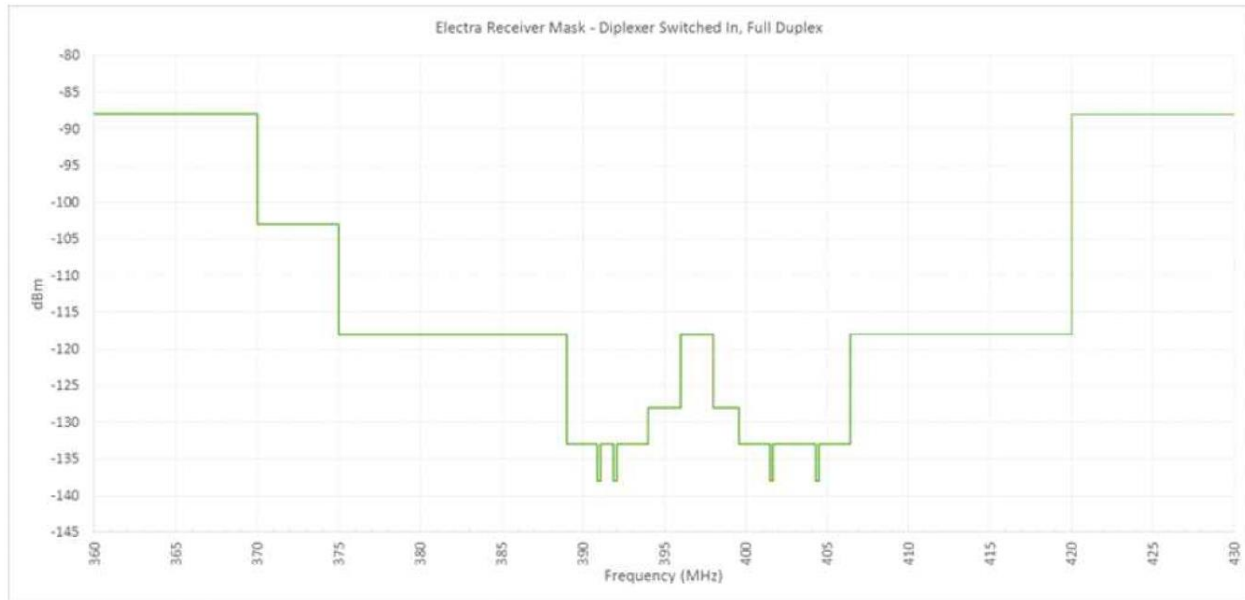
The remainder of this document provides a high-level description of the Electra interfaces to assist the Contractor in incorporating the Electra into their orbiter design.

EMI/EMC Considerations

Due to the relatively low frequencies and typically broad UHF antenna patterns, Electromagnetic Interference (EMI) and Electromagnetic Compatibility (EMC) issues should be carefully considered in accommodation of the Electra on a Mars orbiter. Just as spacecraft subsystems and payloads must be designed and implemented to avoid generation of EMI that would degrade communications on the deep space link to/from Earth, a Mars orbiter carrying the Electra must also ensure mitigation of any EMI that would degrade Electra relay communications at UHF frequencies.

As a reference, Figure 2 shows the EMI levels that Electra can tolerate without degradation across the UHF band. Any interference that couples into the UHF antenna and exceeds these levels into the Electra antenna port may increase receiver error rates or even cause a false carrier lock. Notches are included for the four most common return link channels

Early attention to EMI/EMC considerations in component and spacecraft design, and thorough component-level and system-level EMI/EMC testing, are key to achieving electromagnetic compatibility.



Frequency Range	EMI Threshold (dBm) at EUT Antenna Port
360-370 MHz	-88
370-375 MHz	-103
375-389 MHz	-118
389-390.9 MHz	-133
390.9-391.1 MHz	-138
391.1-391.9 MHz	-133
391.9-392.1 MHz	-138
391.2-394 MHz	-133
394-396 MHz	-128
396-398 MHz	-118
398-399.585625 MHz	-128
399.585625-401.485625 MHz	-133
401.485625-401.685625 MHz	-138
401.685625-404.3 MHz	-133
404.3-404.5 MHz	-138
404.5-406.4 MHz	-133
406.4-420 MHz	-118
420-430 MHz	-88

Figure 2: EUT EMI Thresholds

Electra UHF Transceiver Description

The EUT is a fully-reconfigurable software-defined radio with frequency-agile operation over the 390-450 MHz band in a half-duplex mode and over the respective forward and return sub-bands in a full-duplex mode. New features available in the Electra units for MTN include support for two simultaneous Proximity-1 relay sessions via FDMA, concurrent open-loop recording and closed-loop demodulation, and higher sampling rate open-loop recordings. The Electra uses high-reliability parts, compliant to JPL Parts Engineering Technical Standard (PETS) 2+, and has passed reliability analyses with a specified 8.5-year lifetime. Key EUT specifications are provided in Table 2.

Table 2 Key EUT Specifications

Parameter	Electra UHF Transceiver
Duplex modes	Full Duplex (FD), Half Duplex (HD)
Transmit (TX) Frequency	FD: 435 – 450 MHz HD: 390 – 450 MHz
Receive (RX) Frequency	FD: 390 - 405 MHz HD: 390 – 450 MHz
Operational Modes	Standby, RX, TX, RX/TX, FDMA RX/TX with Two Users
TX/RX Rate	1,2,4,8,..., 4096 kbps
Modulation	Manchester, NRZ-L, BPSK Mod Index 60 or 90 deg
Coding	K=7, R=1/2 Convolutional Encoder/Decoder R=1/2 Low-Density Parity Check Decoder
Spectrum Record	Open Loop Recording Up to 1.125 Msps , 8 bits/sample
RX Noise Figure	3.4 dB (typical)
RF TX Power	>5.0 W
Communications Protocol	CCSDS Proximity-1 Space Link Protocol
Reprogrammable Software/Firmware	Available in-flight with non-volatile memory for multiple back-up images
Radiometric Tracking	Carrier Phase/Doppler, Amplitude; Coherent or Non-Coherent modes
Mass	5.6 kg (w/ internal EMXO)
Dimensions (l,w,h)	26.4 x 20.2 x 15.3
EOL DC Power Over Temperature (28 V input supply voltage)	Standby: 22 W Receive-Only: 29 W Transmit and Receive: 77 W
Parts Grade	JPL PETS 2+
TID	20 krad

The EUT incorporates a modular design with functional elements residing in four stacked modules: a Filtering and Switching Unit (FSU) slice, a Receiver/Modulator

(RXA/MOD) slice, a Digital Processing Module (DPM) slice, and a Power Amplifier-Power Supply Module (PA/PSM) slice. The DPM contains an internal Evacuated Miniature Crystal Oscillator (EMXO, Microchip EX-219). A simplified block diagram of the Electra and its interfaces is shown in Figure 3 and additional details on the digital and power interfaces is provided in Table 3.

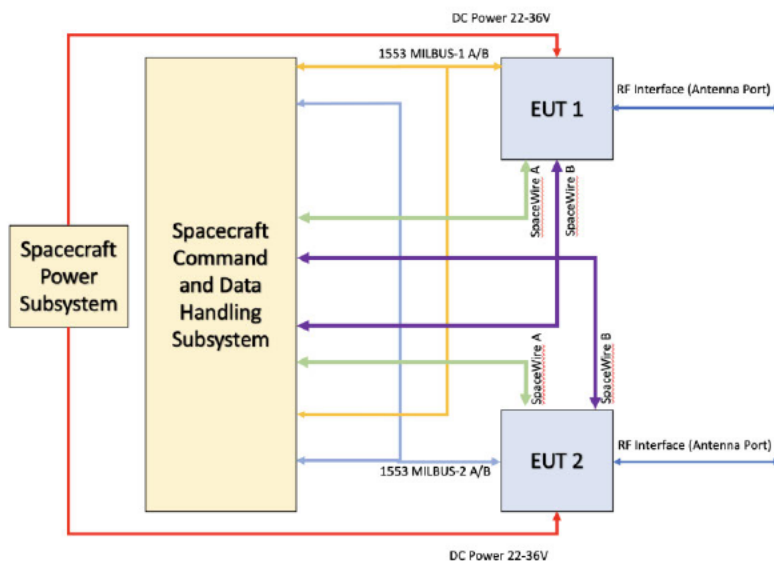


Figure 3: Electra Spacecraft Interfaces (for system with redundant Electra units)

Table 3 Electra – Spacecraft Bus Interfaces

Electra Interface	Description
Command/Control and Telemetry	Redundant 1553-B interface with time transfer via Mode Code 17
Proximity link data (Forward and Return)	Redundant SpaceWire with 10 Mbps forward rate and 24 Mbps return rate
DC power	Isolated input with 22-36 V operating range. No damage 0-40 V

Deliverables and Schedule

For use in the Mars Telecom Network (MTN) Orbiter, NASA will provide specific Government Furnished Equipment (GFE) deliverable items to the selected MTN contractor, as identified in Table 5 in accordance with NFS 1852.245-76 List of Government Property Furnished Pursuant to FAR 52.245-1 (Deviation) (Dec 2025). ...

Table 5 Government Furnished Property (GFP) for MTN

Description	Part number	Manufacturer	Quantity	Unit Acquisition Cost	Available Date	Provided As Is (Yes or No)
Electra EUT FM Radio	E699695-501	L3Harris/JPL	1	\$4.028M	Feb 2027	Yes
Electra EUT FM Radio	E699695-501	L3Harris/JPL	1	\$2.171M	Feb 2027	Yes
Electra EUT PM Baseband Radio	180137	JPL	1	\$0.250M	Oct 2026	Yes

Each item of Electra flight hardware units will be delivered with an End Item Data Package documenting all transceiver level test results. For the EUT, these include functional, environmental, and Proximity-1 protocol compliance testing.